

Soumyadeep Nath

Aspiring Data Scientist & AI Builder | Python · SQL · ML · GenAI · Full-Stack AI

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PROFESSIONAL SUMMARY

From literature to data — building the bridge deliberately. Currently completing an Executive PG Programme in Data Science & AI at IIT Roorkee, working hands-on with Python, SQL, Machine Learning, and Generative AI systems including RAG pipelines, embeddings, and LLM APIs. Built and deployed two live AI applications: **AlignResume**, a production-grade full-stack AI platform with truthfulness guardrails and structured JSON prompt pipelines, and **Future Fit**, a Streamlit-based skill trend analytics platform built on real job market data. A background in academic writing and language analysis provides a unique edge: translating complex model outputs into clear, compelling narratives that stakeholders understand.

EDUCATION & TRAINING

Executive PG Programme in Data Science & AI – In Progress

Feb 2026 – Present

Indian Institute of Technology (IIT) Roorkee | Remote

Coursework: Python · SQL · NumPy · Pandas · Machine Learning · Statistics · Data Visualization · Generative AI · Prompt Engineering · RAG · LLM APIs · Embeddings · Vector Databases

- Engineered data pipelines and automated ETL workflows using Python (Pandas, NumPy) to clean, transform, and prepare structured and unstructured datasets for downstream analysis.
- Queried relational databases using advanced SQL — window functions, joins, subqueries, and aggregations — extracting actionable, analysis-ready business insights.
- Constructed EDA frameworks applying statistical reasoning, hypothesis testing, and A/B testing methodology to validate data-driven decisions.
- Developed and evaluated supervised Machine Learning models — regression, classification, and tree-based ensembles (Random Forest, Gradient Boosting) — using scikit-learn.
- Implemented Generative AI applications by integrating LLM APIs, engineering structured prompts, and building RAG systems with embeddings and FAISS/Chroma vector databases.
- Translated complex model outputs into structured data narratives and interactive dashboards using Matplotlib, Seaborn, and Plotly; documented findings in reproducible Jupyter notebooks.

M.A. in English Literature — First Division

Oct 2022 – Jul 2024

Presidency University, Kolkata

B.A. Honours in English Literature — First Division

Sep 2019 – Jul 2022

Presidency University, Kolkata

PROJECTS

AlignResume — AI-Powered Resume Optimization Platform

May 2026

Live: align-resume-beta.vercel.app · GitHub: github.com/sdn9300/align-resume · Stack: Next.js · TypeScript · Tailwind CSS · Groq API (LLaMA 3.3 70B) · Playwright · Zod · Vercel

- Architected and deployed an end-to-end full-stack AI web application that parses resumes and job descriptions, scores alignment quality, and generates ATS-optimized bullet rewrites via a structured JSON prompt pipeline over the Groq API — with Zod schema validation ensuring deterministic, parseable LLM outputs.
- Implemented an automated truthfulness guardrail layer that validates rewritten bullets against the original resume, blocking export if hallucinated metrics, employers, or skills are detected — prioritizing honesty over inflation.
- Engineered Playwright-based headless PDF export, rate-limited API proxy (10 runs/hr per IP), and GitHub-Vercel CI/CD auto-deploy; deepened expertise in LLM application architecture, prompt engineering, and production-grade full-stack AI deployment.

Future Fit — AI-Powered Skill Trend Intelligence Platform

Jun 2026

Live: skilltrendanalysis-rf3zefgsjaa4l9f8pu2prb.streamlit.app · GitHub: github.com/sdn9300/Future-Fit-AI-Powered-Skill-Trend-Analysis

Stack: Python · Pandas · Plotly · Streamlit · Prophet · mlxtend (Apriori) · SciPy · Groq API (LLaMA 3.3 70B)

- Built an end-to-end Time Series and NLP analysis platform processing 224,605 skill mentions extracted from 50,000+ global AI/Data Science job postings (2020–2026), using a taxonomy-driven regex extraction engine with word-boundary matching

across 100+ skills.

- Engineered a 10-step deterministic cleaning pipeline and developed a Market Basket Analysis (Apriori) engine mapping 230 frequent skill itemsets and extracting association rules (lift ≥ 0.90 , confidence ≥ 0.38) to identify strong co-occurrence trends like PyTorch $\rightarrow$ TensorFlow.
- Implemented a tiered yearly forecasting model using Prophet to project demand shares into 2027, integrating a dynamic fallback to linear trend extrapolation if the 80% confidence interval width exceeded a $1.5\times$ variance threshold.
- Deployed a 6-panel Streamlit dashboard featuring Plotly charts, interactive association rules, and a Groq LLM-powered Skill Gap Advisor enriched with association rule insights to generate contextual learning paths.

Gleaner — Multi-Board Job Scraper & ETL Pipeline

Jun 2026

GitHub: github.com/sdn9300/gleaner-job-scout

Stack: Python · BeautifulSoup4 · Firecrawl API · Google Sheets API · YAML · Git

- Architected and implemented a modular ETL pipeline using the Adapter Design Pattern to concurrently fetch, normalize, and aggregate job listings from 4+ major platforms (Naukri, Indeed, RemoteOK, Wellfound/WWR).
- Engineered robust scraping mechanics using stealth configuration headers, custom bot user-agents, and Firecrawl API's structured extraction logic to ensure reliable, anti-bot-resistant data retrieval.
- Designed data processing stages utilizing exact and similarity-based deduplication algorithms, role-relevance keyword analysis, and location-based filters to clean raw unstructured payloads.
- Integrated automated reporting channels by establishing programmatic write pathways to local CSV outputs and secure synchronization with cloud-hosted spreadsheets using the Google Sheets API (gsread / google-auth).
- Developed a configurable CLI engine driven by centralized YAML files, allowing granular execution control over logging levels, output size limits, and targeted source execution.

SKILLS

Languages	Python, SQL, TypeScript
Frameworks & Tools	Next.js, Tailwind CSS, Streamlit, Vercel, Playwright, Git / GitHub, Firecrawl API, YAML
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, SciPy, Prophet, mlxtend, Zod, BeautifulSoup4, gsread
Machine Learning	Regression, Classification, Random Forest, Gradient Boosting, Model Evaluation, Time Series Forecasting (Prophet), Association Rule Mining (Apriori / mlxtend)
Generative AI	LLM APIs (Groq), Prompt Engineering, Structured JSON Pipelines, RAG, Embeddings, Vector Databases (FAISS / Chroma), Guardrail Design
Data Analytics	EDA, Data Cleaning, ETL Pipelines, Web Scraping, Statistical Analysis, Hypothesis Testing, A/B Testing, Data Storytelling
Visualization	Matplotlib, Seaborn, Plotly, Jupyter Notebook
Soft Skills	Critical Thinking, Analytical Reasoning, Communication, Attention to Detail, Adaptability

CERTIFICATIONS

Machine Learning Statistical Foundations Professional Certificate

LinkedIn Learning × Wolfram Research | Completed: Mar 2026 | Skills: Machine Learning · Linear Algebra · Wolfram Language

Anaconda Python for Data Science Professional Certificate

LinkedIn Learning × Anaconda | Completed: Mar 2026 | Skills: Python · Data Analysis · Statistics

Career Essentials in Generative AI

LinkedIn Learning × Microsoft | Completed: Mar 2026 | Skills: Generative AI · Microsoft Copilot · Responsible AI